

2023 AMC 10A Problem 13 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 13. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 13

Problem:

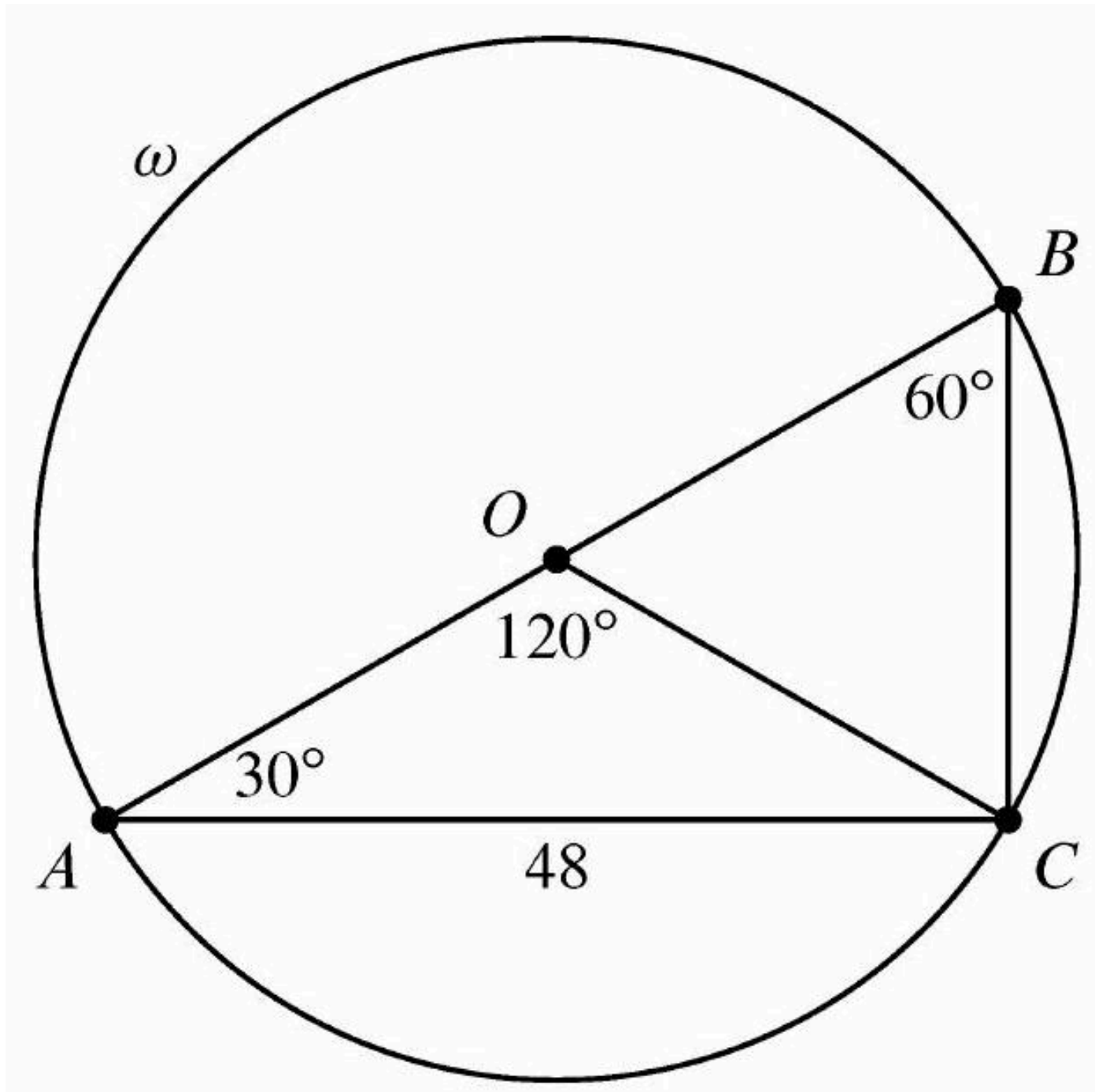
Abdul and Chiang are standing 48 feet apart in a field. Bharat is standing in the same field as far from Abdul as possible so that the angle formed by his lines of sight to Abdul and Chiang measures 60° . What is the square of the distance (in feet) between Abdul and Bharat?

Answer Choices:

- A. 1728
- B. 2601
- C. 3072
- D. 4608
- E. 6912

Solution:

In the diagram below-an overhead view-points A , B , and C represent the locations of Abdul, Bharat, and Chiang, respectively. Let ω be the circle containing A , B , and C , and let O be center of ω . Because $\angle ABC = 60^\circ$, it follows that $\angle AOC = 120^\circ$.



Because B is as far as possible from A , it follows that B must be at the other end of the diameter of ω containing A , as shown.

It remains to calculate AB . Because $\triangle ABC$ is a $30 - 60 - 90^\circ$ right triangle and $AC = 48$, it follows that $AB = \frac{96}{\sqrt{3}}$. Then

$$AB^2 = \frac{96^2}{3} = (\text{C}) \boxed{3072}$$

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